
EEE 6778. Applied Machine Learning II



Module 5. Multimodal Architectures

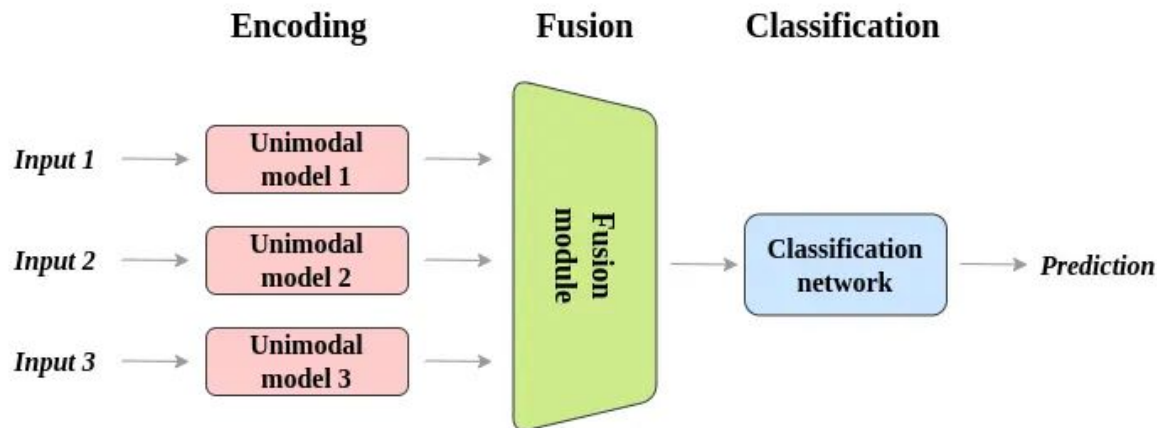
Introduction

- Multimodal learning refers to training models on multiple types of input data (modalities)
- Common modalities: text, images, audio, video, 3D data, sensor data → Multimodal datasets
- Goal: Build models capable of cross-modal understanding and richer representation

This is crucial in real-world tasks where relying on a single data source is insufficient—such as autonomous vehicles using vision, radar, and LiDAR together.

Common Architectures

- Multimodal architectures usually consist of three parts:
 - **Unimodal encoders** that encode individual modalities. Usually, one for each input modality.
 - **A fusion network** that combines the features extracted from each input modality, during the encoding phase.
 - A classifier that accepts the fused data and makes predictions.



Encoding (Representation)

- Multimodal representation is the task of encoding data from multiple modalities in the form of a vector or tensor.
- Good representations that capture semantic information of raw data are very important for the success of deep learning models.
- However, feature extraction from heterogeneous data in a way that exploits the synergies between them is very hard

Joint representation

Each individual modality is encoded and then placed into a mutual high dimensional space. This is the most direct way and may work well when modalities are of similar nature.

Coordinated representation

Each individual modality is encoded irrespective of one another, but their representations are then coordinated by imposing a restriction.

Encoding- Example

- **Scenario:** You're working with a multimodal model that takes both images and text as inputs — say, product photos and their descriptions for an e-commerce recommendation system.

Encoding Individually: The image is passed through a CNN to get an image embedding. The text description is passed through a Transformer or BERT to get a text embedding. These are two completely separate pipelines — they don't interact during this encoding phase.

Coordination via Restriction: After encoding, the model forces the embeddings to lie in a shared space (e.g., via contrastive learning, minimizing distance between matching image-text pairs while pushing non-matching pairs apart). This restriction coordinates the two modalities — though they were encoded independently, the shared embedding space links their semantics.

Fusion Strategies

- The method/architecture used for fusion is probably the most important ingredient for success.
 - **Early Fusion:** Combine raw or low-level features from modalities
 - Example: concatenating image pixels and text embeddings
 - **Late Fusion:** Use separate models and combine predictions
 - Example: voting or averaging outputs
 - **Hybrid Fusion:** Integrate modalities at multiple layers (hierarchical or adaptive)

Fusion Strategies

- The simplest method is to use simple operations such as concatenating or summing the different unimodal representations.
- However, more sophisticated and successful methods have been researched and implemented.
 - For example, the cross-attention layer mechanism is one of the more recent and successful fusion methods. It has been used to capture cross-modal interactions and fuse modalities in a more meaningful way.

$$\alpha_{kl} = s\left(\frac{K_l Q_k}{\sqrt{d}}\right) V_l$$

Fusion Strategies

- Cross-Attention (Recap)

$$\alpha_{kl} = s\left(\frac{K_l Q_k}{\sqrt{d}}\right) V_l$$

- Where a_{kl} denotes the attention score vector, s denotes the softmax function, K , Q , and V are the Key, Query and Value matrices of the attention mechanism respectively. For symmetry is also computed, and the two may be summed up to create an attention vector that maps the synergy between the two modalities involved.

Fusion Strategies

- Cross-Attention (Recap)

$$\alpha_{kl} = s\left(\frac{K_l Q_k}{\sqrt{d}}\right) V_l$$

- In the case of **three or more modalities**, multiple cross-attention mechanisms may be used so that every different combination is calculated.
 - For example, if we have vision (V), text (T), and audio (A) modalities, then we create the combinations VT, VA, TA, and AVT in order to capture all possible cross-modal interactions.
- Even after using an attention mechanism, a concatenation of the above cross-modal vectors is often performed to produce the fused vector F .

Classification or Regression

- Once the fusion has been completed, vector F is fed into a classification/regression model.
- This is usually a feedforward network.
- The input vector F encodes complementary information from multiple modalities, thus providing a richer representation compared to the individual modalities V , A , and T .

**Can you think on
regression
examples?**

Common Architectures

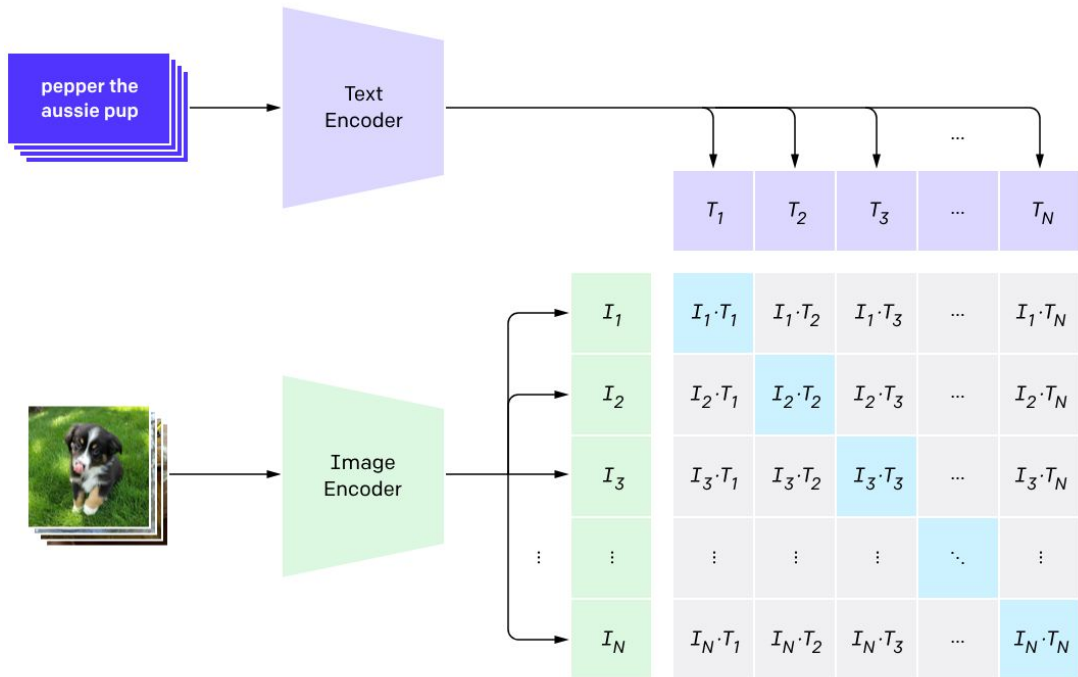
- **Dual Encoder:**
 - Each modality has its own encoder, trained separately.
 - **Fusion:** Happens after encoding, often using dot product or cosine similarity.
 - **Use Case:** Retrieval tasks (e.g., find image that matches a caption)
 - **Example:** CLIP (Contrastive Language–Image Pretraining) Image → Vision Transformer *and* Text → Text Transformer
 - Both mapped to a shared embedding space
 - Matched using **contrastive loss**



Common Architectures

- Dual Encoder - CLIP architecture

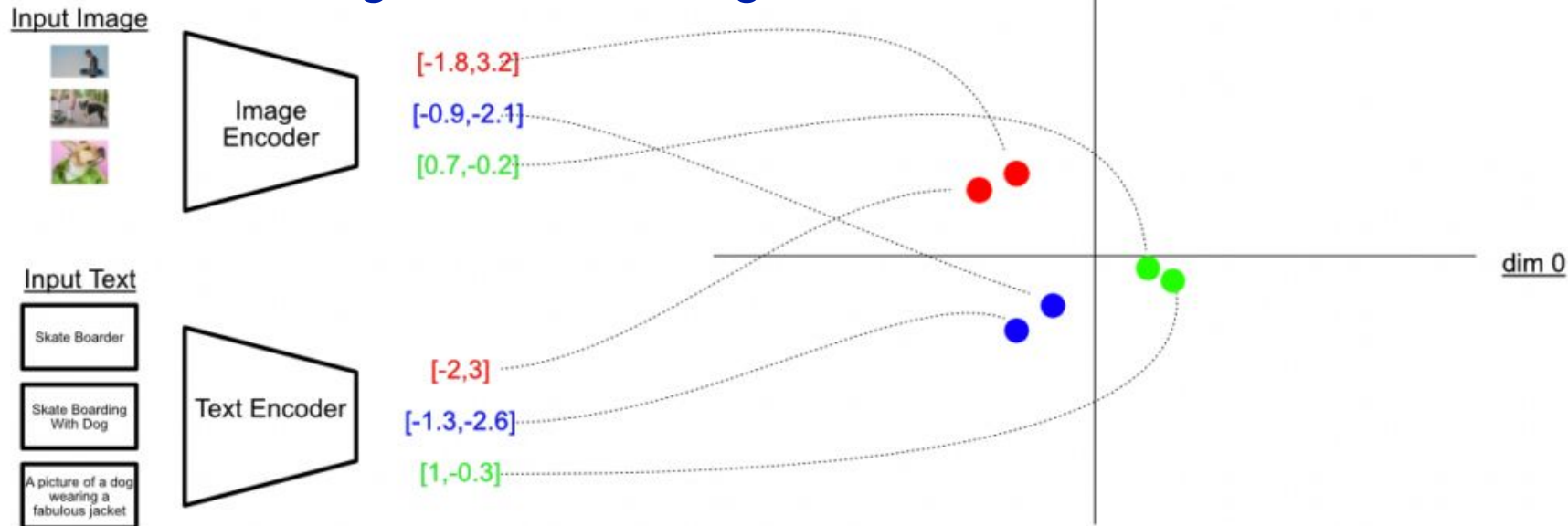
1. Contrastive pre-training



Common Architectures

- Dual Encoder - CLIP architecture

By putting similar images and text in *similar* spots, and different images and text in different spots, CLIP style models learn to make decisions about which text belongs with which image.

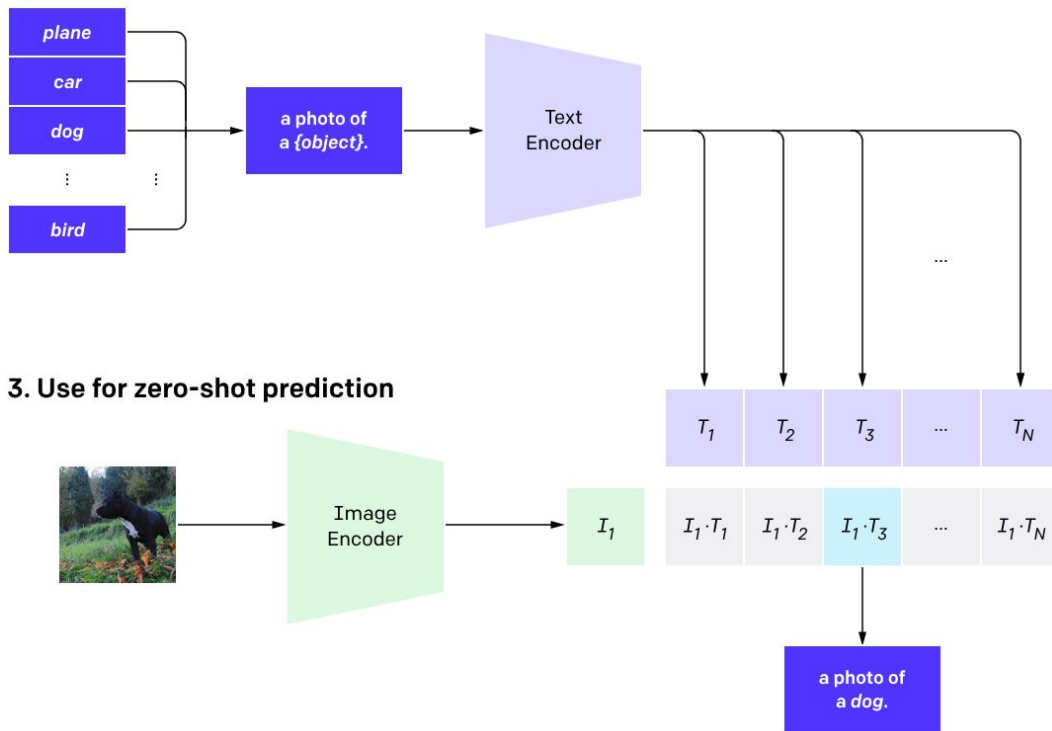


Common Architectures

- Dual Encoder - CLIP architecture

All of a dataset's classes are converted into captions such as "a photo of a dog" and predict the class of the caption CLIP estimates best pairs with a given image.

2. Create dataset classifier from label text



Common Architectures

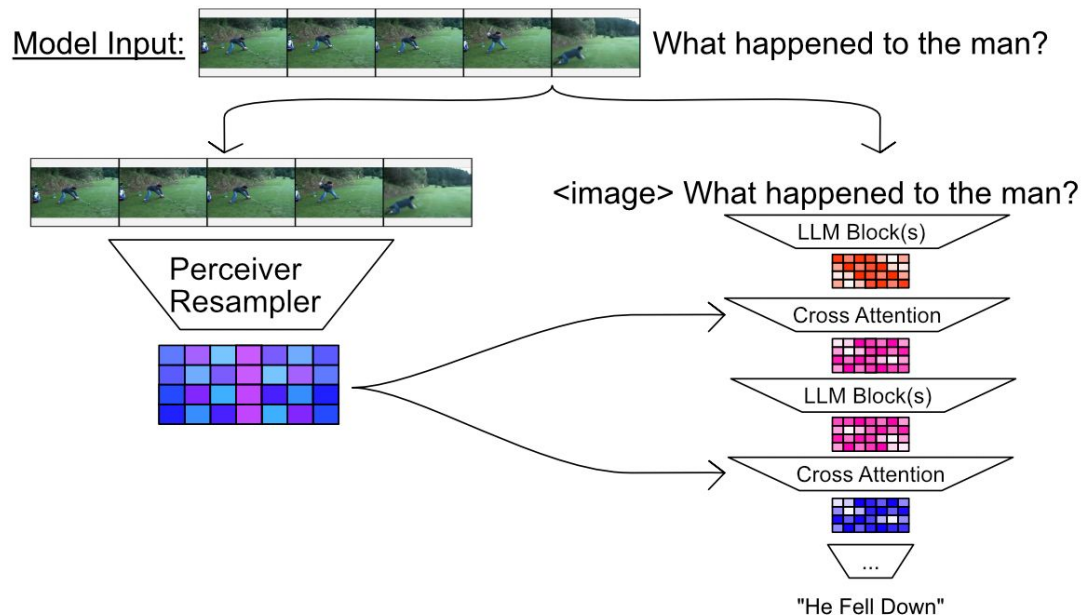
- **Cross-Attention Models**

- Use Transformers that allow one modality to attend to another.
- **Mechanism:** One stream acts as the query, the other as key/value.
- **Use Case:** Tasks like visual question answering or **few-shot** image captioning
- **Example:** Flamingo Visual Language Model
 - Cross-attention layers where text attends to image features. Enables fast adaptation with few training samples



Common Architectures

- Cross-Attention Models - Flamingo



Common Tasks

- Image Captioning
- Image Retrieval
- Text-to-image Generation
- Visual Question Answering
- etc



A horse carrying a large load of hay and two people sitting on it.



Bunk bed with a narrow shelf sitting underneath it.



No people and
switch to night-time

Giant cobra
snake made
of corn



Applications Across Domains - Use Cases

- **Healthcare:** Imaging + EHR + genomics (e.g., cancer diagnosis)
- **Autonomous Driving:** Camera + LiDAR + radar for scene understanding
- **Robotics:** Vision + touch + audio for object manipulation
- **Media & Art:** Text-to-image generation (e.g., Stable Diffusion)
- **HCI:** Speech + gesture + facial recognition in smart assistants

Multimodal Datasets

- **Here are some key datasets:**
 - **COCO-Captions:** 330K images with text descriptions; used for image captioning.
 - **VQA:** 265K images with visual questions; useful for visual question answering and image understanding.
 - **CMU-MOSEI:** 23.5K video clips with audio and text for emotion and sentiment analysis.
 - **Social-IQ:** 1,250 annotated videos focused on social interaction, reasoning, and Q&A.
 - **Kinetics (400/600/700):** YouTube video clips with actions and sound; used for action recognition.
 - **RGB-D Object Dataset:** 250K RGB and depth images of 300 objects for 3D object detection and depth tasks.

Core Technical Challenges

- **Representation:** Extracting meaningful embeddings from each modality
- **Alignment:** Mapping features from different modalities to a common space
 - Audio and video data are often asynchronous. Aligning them requires temporal attention mechanisms
- **Fusion:** Combining multimodal features meaningfully
 - Can occur at multiple levels and must be designed based on task-specific demands.
- **Co-learning:** Sharing knowledge across modalities
- **Decoding:** Generating coherent unimodal or multimodal outputs